Class 19

Suppose Γ is an extensive form game we can write it as a normal form game

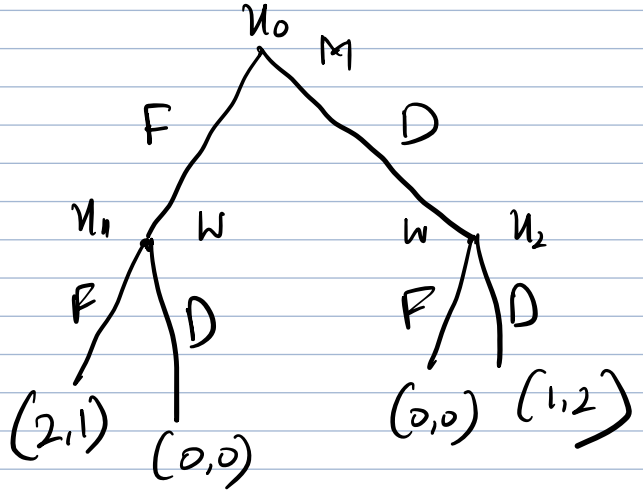
$$\Gamma = \langle N, V, E, v_0, (V_i)_{i \in N}, (U_i)_{i \in N} \rangle$$

For the normal form game

$$\langle N, (S_i)_{i \in N}, (\hat{U}_i)_{i \in N} \rangle$$

$S_i : V_i \to \bigcup_{v \in V_i} A(v)$ s.t. $S_i(v) \in A(v)$ $\forall v \in V_i$

- N is the set of players in the extensive form game
- S_i is the set of strategies for i-th player in the extensive form game.
- $\hat{U}_i(s) = U_i(\bar{v})$ where the strategy profile s uniquely corresponds to the leaf $\bar{v}$.

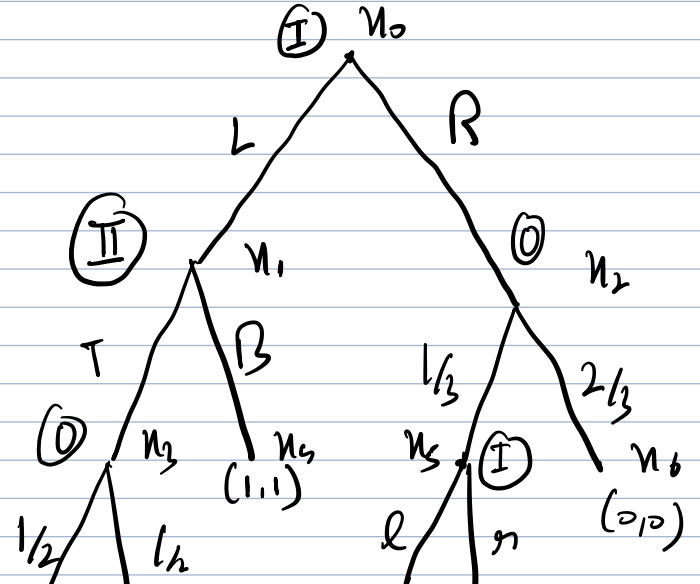


$N = \{M, W\}$ $S_1 = \{F, D\}$ $S_2 = \{(F,F), (F,D), (D,F), (D,D)\}$

	(F,F)	(F,D)	(D,F)	(D,D)
F	(2,1)	(2,1)	(0,0)	(0,0)
D	(0,0)	(1,2)	(0,0)	(1,2)

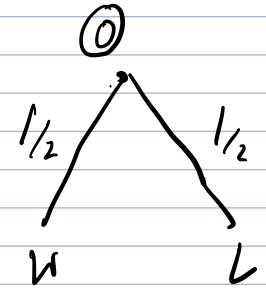
$$\Gamma = \langle N, V, E, v_0, (V_i)_{i \in N \cup \{0\}}, (p_v)_{v \in V_0}, (u_i)_{i \in N} \rangle$$

- N
- (V, E, v_0)
- $(V_i)_{i \in N \cup \{0\}}$: a partition of the non-leaf nodes among the agents and the nature where 0 corresponds to nature
- $(p_v)_{v \in V_0}$: a set of prob distributions for each $v \in V_0$ over the edges coming out of v, i.e. $A(v)$
- $(u_i)_{i \in m}$: utility functions defined at the leaf nodes.



n_7 n_8 n_9 n_{10}

(1,0) (0,1) (3,2) (1,0)

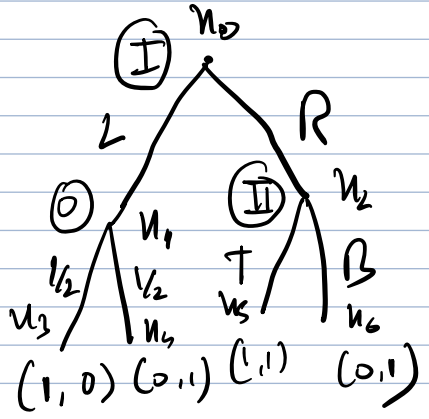


Zermelo or von Neumann's theorem does not hold.

- Play of the game is same as before. Just that whenever there is a chance node, a lottery takes place and based on the lottery the action is taken at that node.

- strategies are defined in the same way as before

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$$u_1(L,T) = \frac{1}{2}u_1(n_3) + \frac{1}{2}u_1(n_4) = \frac{1}{2}$$

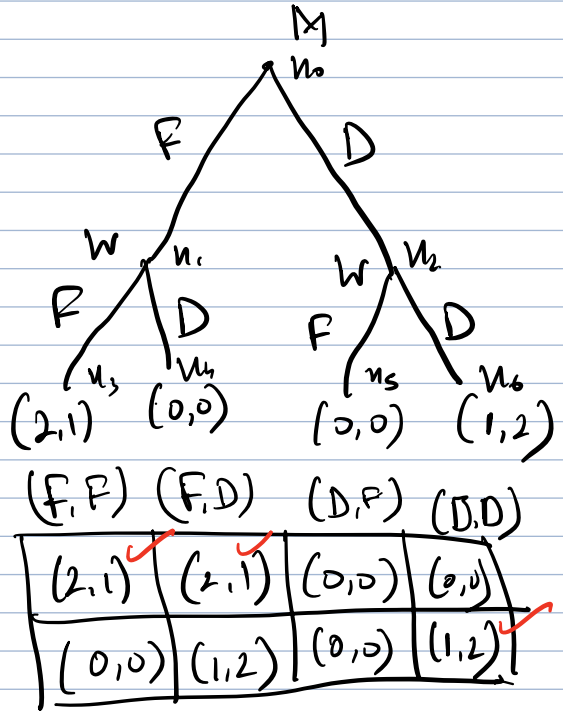
$$u_2(L,T) = \frac{1}{2}u_2(n_3) + \frac{1}{2}u_2(n_4) = \frac{1}{2}.$$

Utilities are calculated as expected utilities according to the prob dist over the end nodes.

Nash Equilibrium

A strategy profile s^* is an NE if

$$u_i(s^*) \geq u_i(s_i, s_{-i}^*) \quad \forall i \in N$$
$$\forall s_i \in S_i$$



	(F,F)	(F,D)	(D,F)	(D,D)
F	(2,1)	(2,1)	(0,0)	(0,0)
D	(0,0)	(1,2)	(0,0)	(1,2)

$(F, (F,F))$ NE — not SPE

$(F, (F,D))$ NE — SPE

$(F, (D,F))$ (not an NE) — not SPE

$(D, (D,D))$ — not a SPE

Subgame perfect Equilibrium

A strategy profile is a subgame perfect Equilibrium (SPE) if and an NE in all the subgames.